AI Space Chameleon — Project Plan

Sep 28, 2026 · @tharun

1. Executive summary

AI Space Chameleon is a one-line embeddable support widget that adapts to any website's look, answers customers with RAG, runs multi-step requests through a data-structure-based task engine ("AI Stacks and Queues"), and hands the chat to a human agent with full context when it cannot finish. The MVP launches on Feb 1, 2027, 17 weeks after kickoff, with a 7-person team and targets small and mid-sized online stores first.

The problem. Support teams are overloaded with repeat questions, and today's AI agents are either expensive per-resolution add-ons locked to one helpdesk (Fin at $0.99, Zendesk at $1.50 to $2.00, Gorgias at $0.90 to $1.00 per resolution) or simple FAQ bots that cannot carry out multi-step tasks reliably. Customers also distrust bots that trap them: a Gartner study found 64% of customers would prefer companies not use AI for service.

The bet. Three things together are the wedge:

1. **Chameleon widget**: one `<script>` tag, auto-themes to the host site, works on any stack (Shopify, WordPress, React, plain HTML).
2. **AI Stacks and Queues**: every request becomes a structured task. A priority queue orders work, a stack breaks a request into ordered sub-steps with rollback, and a dependency graph runs independent steps in parallel. The result is predictable, auditable execution instead of one opaque LLM call.
3. **Honest handoff**: when confidence drops, a step fails, or the customer asks, the task moves to a priority queue of human agents with a summary, the step trail and suggested next action. Customers are never trapped.

Headline targets for MVP (6 months after launch):

Metric	Target
Automated resolution rate	40% of chats
Time to install	under 10 minutes
Median first response	under 3 s
Handoff wait (business hours)	under 2 min
Paying customers	50 stores

Metric	Target
Hallucination rate on eval set	under 2%

Ask: approve a 2-week discovery and proof of concept, then the MVP build described in section 9.

2. Market research

The market is large and growing about 20% a year, but pricing is the top buyer complaint: per-resolution fees make the best months the most expensive, and most AI agents only work inside their own helpdesk. That leaves room for a flat-priced, helpdesk-neutral widget that executes tasks reliably.

Market size

- Grand View Research puts the global chatbot market at $9.6B in 2025 and $11.8B in 2026, reaching $41.2B by 2033 (19.6% CAGR). Customer service is the largest application and retail/e-commerce the largest vertical. (source)
- MarketsandMarkets sizes the wider AI-for-customer-service market at $12.06B in 2024, rising to $47.82B by 2030 (25.8% CAGR). (source)
- Gartner predicts agentic AI will resolve 80% of common service issues without a human by 2029, cutting operating costs about 30%. (source)
- The counterweight: Gartner also predicts half of companies that cut service staff because of AI will rehire by 2027, and only 20% of service leaders surveyed had actually reduced staff. Human handoff stays essential. (source)

Serviceable market for the MVP (estimate, to validate in discovery): English-speaking online stores and SaaS companies with 2 to 50 support agents that run their own website. These buyers want AI now but balk at enterprise suites.

Competitors

Product	Pricing model (2026)	Strength	Gap we exploit
Fin (formerly Intercom)	$0.99 per outcome + seats from $29/mo; 50-outcome minimum off Intercom	Strong AI, now owned by Salesforce (closed Sep 10, 2026)	Bill grows with success; assumed resolutions billed after 24 h of silence; repackaging risk under Agentforce

Product	Pricing model (2026)	Strength	Gap we exploit
Zendesk AI agents	About $1.50 committed / $2.00 pay-as-you-go per automated resolution, on top of Suite seats	Deep helpdesk	Needs Zendesk Suite; layered add-ons; auto overage billing since Jan 2026
Gorgias AI Agent	$0.90 to $1.00 per resolution, and AI tickets also count as helpdesk tickets	Shopify order actions	Shopify-centric; double billing
Tidio Lyro	Add-on from $39/mo for 50 AI conversations; Plus from $749/mo	Cheap entry, SMB friendly	Three separate quotas; AI capped at low volume
Chatbase	Free, $40, $150, $500/mo credit plans	Fastest setup, actions, handoff	Credit caps (bot goes "unavailable" at zero); branding removal extra; actions capped per plan

Gaps and positioning

1. **Predictable price.** Flat tiers by conversation volume, never per resolution. The customer is not punished for success.
2. **Helpdesk-neutral.** Hand off to Zendesk, Freshdesk, Gorgias, Slack or our own lightweight inbox.
3. **Reliable multi-step tasks.** The task engine shows every step, retries safely, and rolls back half-finished actions. Competitors mostly expose actions as single tool calls.
4. **Looks native.** Auto-theming removes the design work that makes most widgets look bolted on.
5. **Trust by design.** Clear AI disclosure, a "talk to a human" button that always works, and visible sources on every answer.

Positioning line: "The support widget that blends into your site, finishes the job, and knows when to call a human."

3. Product vision and concept

The product has two halves: the **Chameleon widget** that customers see, and the **AI Stacks and Queues engine** that turns each request into a structured, trackable task. Everything the AI does is a task with steps, a state and an owner, so it can be paused,

retried, audited or handed to a person.

The Chameleon widget

- **One-line install:** `<script src="https://cdn.spacechameleon.ai/w.js" data-site="abc123"></script>`, plus plugins for Shopify and WordPress.
- **Auto-theming:** on load it reads the host page's computed colors, fonts and corner radius and builds matching design tokens. The site owner can lock or override them in the dashboard.
- **Isolation:** renders inside Shadow DOM so host CSS cannot break it and it cannot break the host.
- **Small and fast:** loader under 15 KB gzipped; the full chat bundle loads lazily when the bubble is first opened.
- **Context aware:** knows the current page, cart and logged-in customer (through a signed token from the host), so answers fit the moment.

AI Stacks and Queues: how a request runs

Each customer message is classified into an intent. Simple questions are answered directly with RAG. Anything that needs actions ("change my delivery address", "return item 2 and exchange for size M") becomes a **Task** that the engine plans and executes step by step.

Data structure	Where it is used	Why it fits
Priority queue (min-heap)	Incoming task queue, ordered by SLA deadline, customer tier and urgency	Most urgent work always runs first; O(log n) insert and pop
Stack (LIFO)	Plan execution: a task is decomposed into sub-steps pushed onto a stack; nested sub-tasks push on top	Depth-first execution matches how multi-step requests unfold; easy to resume where it paused
Undo stack	Every side-effecting step pushes its compensating action (e.g. "cancel the return label")	If a later step fails, pop and run compensations in reverse to leave no half-done work (saga pattern)
DAG + topological sort	Steps with dependencies (verify identity before address change; independent lookups in parallel)	Guarantees correct order and lets independent steps run concurrently

Data structure	Where it is used	Why it fits
FIFO queue per agent group	Human escalation queue, one per skill (billing, returns, tech)	Fair ordering within a skill; combined with priority for VIP or angry customers
Dead-letter queue	Steps that fail after retries	Nothing is silently lost; ops can inspect and replay
Sliding window / token bucket	Rate limits per site and per visitor	Stops abuse and protects LLM spend
LRU cache	Frequent answers and embeddings	Cuts latency and cost for repeat questions
Finite state machine	Task lifecycle: queued → planning → running → waiting-on-customer → done / failed / escalated	Every task is always in exactly one known state, which the dashboard shows

Worked example: "Return the blue shoes and send size 9 instead." The planner produces four steps: verify order ownership, create return, check size-9 stock, create exchange order. Steps 1 then 2 run in order; stock check runs in parallel with step 2. If stock is out at step 3, the engine pops the undo stack (cancels the return label) and offers the customer a refund or a human. The customer sees a progress card with each step ticked off.

4. Target customers, personas and use cases

The first customer is a small-to-mid online store (2 to 50 agents) whose support is swamped by order-status and return questions. SaaS help desks and service businesses (bookings, clinics) come after the MVP.

Persona	Who they are	Main pain	What wins them
Store owner (buyer)	Runs a Shopify or WooCommerce store, 1 to 5 staff	No time to answer chats; cannot afford per-resolution bills	10-minute install, flat price, visible savings report
Support lead (admin)	Manages 5 to 50 agents	Repeat questions, long queues, no insight into AI mistakes	Task dashboard, escalation queue, review tools

Persona	Who they are	Main pain	What wins them
Support agent (user)	Answers the chats the AI hands over	Customers repeat themselves after handoff	Handoff card with summary, steps done and suggested reply
End customer	Shops on the site	Waiting, being trapped by bots	Instant answers, progress card, human in one tap
Developer (integrator)	Adds the widget and connects APIs	Brittle integrations, CSS clashes	Script tag, Shadow DOM, typed SDK, webhooks

Top use cases for the MVP

1. **Where is my order?** Lookup by signed customer token or order number + email.
2. **Returns and exchanges** as a multi-step task with rollback.
3. **Policy questions** (shipping, returns, warranty) answered from help articles with a source link.
4. **Change or cancel an order** before it ships.
5. **Product questions** (size, stock, compatibility) from the catalog.
6. **Talk to a human** at any time, with context passed along.

5. Product requirements (PRD)

v1 is a web widget in English for online stores, with RAG answers, up to 5 action types run by the task engine, and handoff to a built-in agent inbox or Zendesk.

Scope

In scope (v1)	Out of scope (v1)
Web widget with auto-theming, Shopify + WordPress plugins	Mobile SDKs, WhatsApp, email, voice
RAG over help articles, URLs, PDFs	Training custom models
Task engine with order lookup, return, exchange, address change, cancel	Payments and refunds executed by AI
Built-in agent inbox + Zendesk handoff	Freshdesk, Gorgias, Salesforce connectors (v2)

In scope (v1)	Out of scope (v1)
Admin dashboard: setup, knowledge, tasks, analytics, feedback	Multi-language (v2)

Functional requirements

- FR-1: Widget installs with one script tag and themes itself from the host page.
- FR-2: Every AI answer shows its source article or the data it used.
- FR-3: Action requests become tasks with visible steps; the customer sees a progress card.
- FR-4: Any failed side-effecting step triggers compensation (undo) of earlier steps.
- FR-5: The "Talk to a human" button is always visible and works in one tap.
- FR-6: On handoff, the agent receives a summary, transcript, completed steps and a suggested reply.
- FR-7: Admins can edit knowledge, set tone, pick which actions are enabled and set escalation rules.
- FR-8: Every conversation and task step is logged and searchable.

Non-functional requirements

Area	Requirement
Latency	First token under 1.5 s, full answer under 3 s for 95% of messages
Availability	99.9% monthly for the chat API
Scale	1,000 concurrent chats across tenants at launch
Widget weight	Loader under 15 KB gzipped; no layout shift on host page
Security	Tenant isolation, signed customer tokens, encryption in transit and at rest, PII masking before the LLM
Privacy	GDPR/CCPA data export and deletion; configurable retention
Accessibility	WCAG 2.1 AA (keyboard, screen reader, contrast)

AI requirements

- Answer correctness at least 90% and groundedness at least 98% on the eval set.
- Refuse off-topic, harmful and prompt-injection requests (100% on the red-team set).
- Always disclose the customer is talking to an AI.
- Escalate when confidence is below the tenant's threshold, after 2 failed attempts, or on negative sentiment.

Example user stories

SC-12 Order status. As a logged-in customer, I want to ask "where is my order?" so I get tracking without waiting.

- Given one open order, when I ask, then I see order number, status and tracking link.
- Given several open orders, when I ask, then the widget asks which one.
- Given I am not logged in, when I ask, then it asks for order number and email and reveals nothing until both match.

SC-20 Exchange with rollback. As a customer, I want to exchange an item for another size so I don't need an agent.

- Given the new size is in stock, when I confirm, then a return label and exchange order are created and both steps show as done.
- Given the new size is out of stock, when the check fails, then the return label is cancelled and I am offered a refund or a human.

SC-31 Handoff. As an agent, I want the full context when a chat reaches me so the customer never repeats themselves.

- Given an escalated chat, when I open it, then I see a 3-line summary, the steps completed, the failure reason and a suggested reply.

SC-40 Auto-theme. As a store owner, I want the widget to match my brand without design work.

- Given my site's primary color and font, when the widget loads, then its bubble, header and buttons use them and pass AA contrast.

6. Features, improved and new, each with a feedback loop

Every feature collects a specific signal, and every signal has a named destination (eval set, knowledge base, routing rules or roadmap), so feedback always changes the product instead of sitting in a report. Features improved over the HelpBot reference design are marked (improved); the rest are new.

Feature	What it does	Feedback captured	Where the feedback goes	Release
Answer with sources (improved)	Every answer cites the article or data used	Thumbs up/down + optional reason chips (wrong, outdated, unclear)	Down-voted answers become eval cases; "outdated" flags the article owner	MVP
Task progress card (new)	Shows each step of a multi-step task live	Per-step "this is wrong" tap	Step failure stats tune the planner prompt and action configs	MVP
Undo / rollback (new)	Compensates half-finished actions	Automatic: every rollback logged with cause	Weekly rollback report; top causes become backlog items	MVP
Confidence-based handoff (improved)	Escalates on low confidence, repeated failure or negative sentiment	Agent marks each handoff "needed" or "AI could have handled"	Re-tunes the escalation threshold per tenant	MVP
Agent handoff card (improved)	Summary, steps done, failure reason, suggested reply	Agent rates the summary (1 to 5) and edits the suggested reply	Edited replies become training examples for the summary prompt and new knowledge drafts	MVP
Auto-theming (new)	Matches host site colors and fonts	Owner accepts or overrides tokens in the dashboard	Override patterns improve the theme-extraction heuristics	MVP
Post-chat CSAT (improved)	1 to 5 rating + free text at chat end	Rating and comment	Low-CSAT chats go to the weekly review sample	MVP
Knowledge gap finder (new)	Clusters questions the AI could not answer	Admin approves or dismisses each gap	Approved gaps create draft articles for the owner to publish	v1.1

Feature	What it does	Feedback captured	Where the feedback goes	Release
AI-drafted articles (new)	Turns resolved human chats into draft help articles	Admin edits and publishes	Published drafts feed RAG; edit distance tracks draft quality	v1.1
Proactive nudges (new)	Opens the widget on signals like checkout errors or long idle on the returns page	Nudge dismissed vs. engaged	Nudges with low engagement are auto-paused	v1.1
Agent copilot (new)	Suggests replies inside the agent inbox	Suggestion used, edited or ignored	Usage rate by intent shows where AI is ready to go autonomous	v1.1
Sandbox replay (new)	Replays past conversations against a new prompt or model before release	Side-by-side scores	Release gate: new version must match or beat current scores	v1.1
Feedback inbox for admins (new)	One place for all thumbs-down, low CSAT and agent flags	Triage status (fixed, won't fix, needs article)	Closed items feed the monthly quality report	MVP
Multi-language (new)	Detects language and replies in it	Language-specific thumbs rate	Per-language eval sets; languages below target stay in beta	v2
Omnichannel (new)	Same engine on WhatsApp, email, Instagram	Channel-level CSAT	Channel roadmap priority	v2

How the feedback loop runs

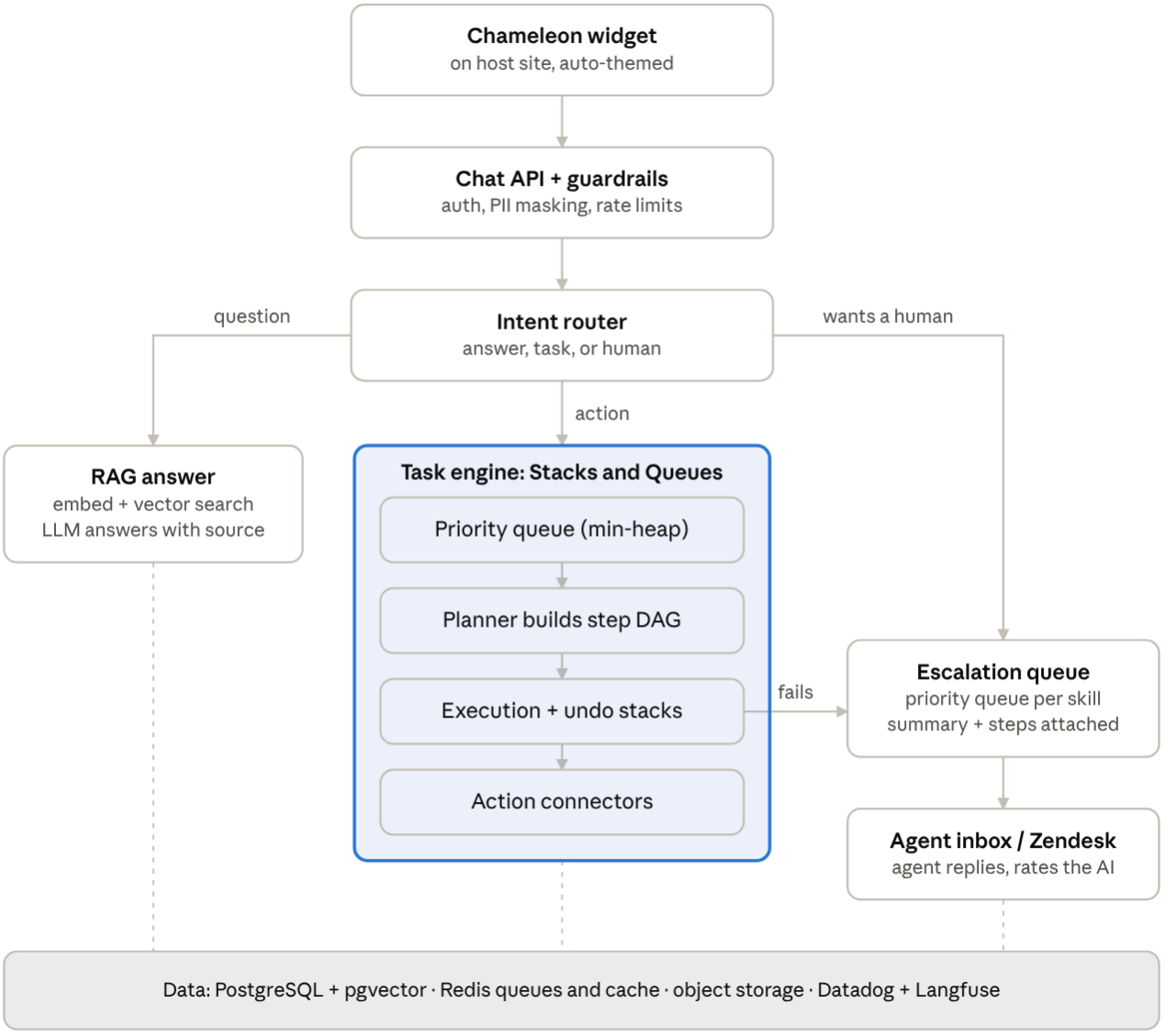
1. **Capture** in the widget, the agent inbox and the dashboard (table above).
2. **Triage** daily: an automatic classifier tags each item (knowledge gap, wrong answer, action failure, UX issue); the support lead reviews a 100-conversation sample weekly.
3. **Act:** wrong answers become eval cases, gaps become articles, action failures become bug tickets, UX issues go to the design backlog.

4. **Verify:** the next release must pass the enlarged eval set; the dashboard shows the before/after for each fixed item.
5. **Close the loop:** customers who left a comment and gave an email get a "we fixed this" note (opt-in).

7. Architecture and tech stack

One multi-tenant backend serves every site: the router sends plain questions to RAG, action requests to the task engine, and "human please" straight to the escalation queue. The task engine is the differentiator; its queues live in Redis and its task records in PostgreSQL so nothing is lost on restart.

One router sends each request to RAG, the task engine, or a human



system architecture · 3 paths from the intent router

A failed step in the task engine first runs the undo stack, then moves the task to the escalation queue with its trail attached.

Core data model

- `tenants`, `sites` (theme tokens, settings), `users` (admins, agents)
- `conversations`, `messages`, `feedback`
- `tasks` (state, priority, deadline, owner), `task_steps` (order, depends_on, status, compensation), `dead_letters`
- `documents`, `document_chunks` (pgvector column), `eval_cases`

Tech stack

Layer	Choice	Why
Widget	Preact + TypeScript, Vite, Shadow DOM	Tiny bundle, isolated styles
Dashboard + agent inbox	React + TypeScript, Tailwind	Fast to build, one design system
API	Python 3.12 + FastAPI, WebSockets / SSE	Streaming answers; Python is standard for AI work
Task workers	Python workers on Redis Streams (or Celery)	Priority queues, retries, consumer groups
LLM	Hosted LLM API with a small model for routing and a larger one for planning	Cost control; swap models behind one interface
Retrieval	Embedding model + PostgreSQL pgvector	One database for data and vectors
Cache and rate limits	Redis	LRU cache, token buckets, sessions
Hosting	Docker on AWS (ECS Fargate for MVP, EKS later), CloudFront CDN for the widget	Low ops for a small team; CDN keeps the widget fast
IaC and secrets	Terraform, AWS Secrets Manager	Repeatable environments; no keys in code
Observability	Datadog or Grafana + Sentry; Langfuse for LLM traces	Latency, errors, token cost, answer quality

Layer	Choice	Why
Integrations	Shopify Admin API, WooCommerce REST, Zendesk API, webhooks	Covers the first customer segment

8. AI-to-human escalation design

The AI hands over whenever it cannot finish safely, and the customer never has to repeat themselves: the agent gets the summary, the steps already done and a suggested reply. Escalations with no outcome are free to the tenant, so we are never paid to keep customers away from humans.

When the AI escalates

Trigger	Rule (tenant-configurable default)	Priority boost
Customer asks for a human	Any wording, or the button	Normal
Low confidence	Answer confidence below 0.6, or no source found	Normal
Repeated failure	2 failed attempts at the same intent	+1
Task step failure	A step fails after 3 retries and rollback completes	+1
Negative sentiment	Frustration detected in 2 consecutive messages	+2
Sensitive topic	Legal threats, safety, payment disputes, account security	+2, routed to a lead
VIP customer	Tenant marks a customer tier as VIP	+1

How a handoff runs

1. The engine freezes the task and writes a **handoff packet**: 3-line summary, intent, customer details the tenant allows, completed and rolled-back steps, failure reason, suggested reply.
2. The packet enters the **escalation priority queue** for the right skill group. Order = priority first, then time waiting.

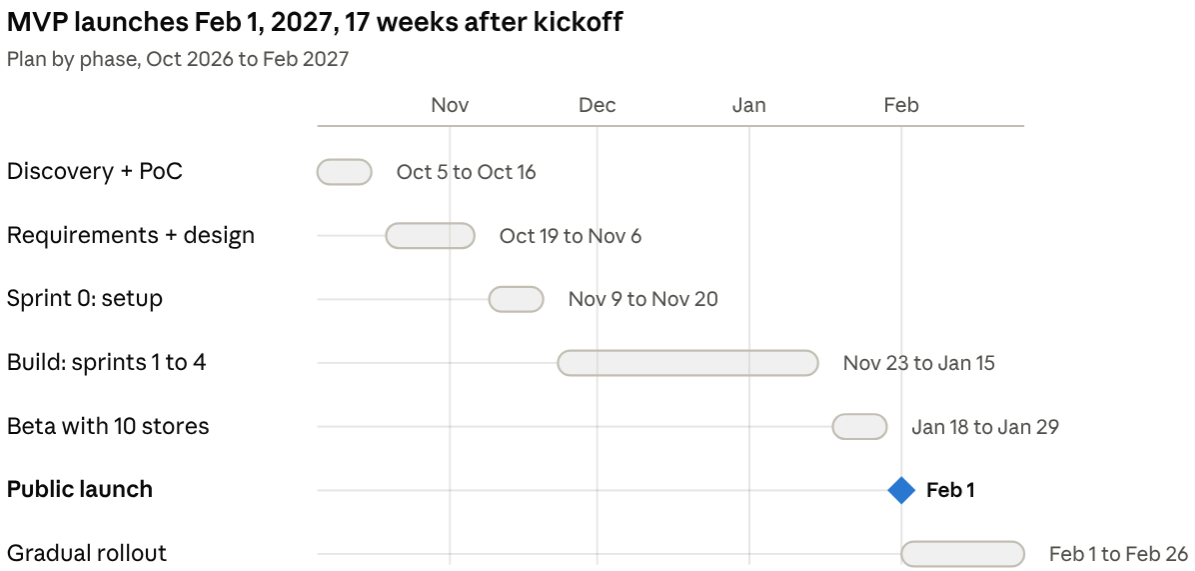
3. The widget tells the customer honestly: "I'm connecting you to a person. Current wait is about 2 minutes." The position updates live.
4. **Agent available:** the first free agent in that group claims the chat (claim is atomic, so two agents never take the same chat). The agent sees the packet and replies in the same widget thread.
5. **No agent online / after hours:** the customer can leave an email; the chat becomes a ticket in the built-in inbox or Zendesk with the packet attached, and the customer gets a reference number.
6. **Wait too long:** if the wait passes the tenant's SLA (default 5 minutes), the item gains priority and the lead is notified.
7. **After the human resolves it**, the agent marks "AI should have handled this" or "needed a human". Those labels retune thresholds and add eval cases (section 6).

Agent experience

- Inbox with queues per skill, wait timers and priority badges.
- One-click "send suggested reply", editable before sending.
- "Hand back to AI" for the easy tail of a chat (for example, sending a tracking link after the agent approved an exception).
- Internal notes and transfer to another group with the packet kept.

9. MVP plan: scope, team, sprints

A 7-person team builds the MVP in five two-week sprints after 5 weeks of discovery and design, runs a 2-week beta with 10 stores, and launches on Feb 1, 2027. Dates assume a kickoff on Oct 5, 2026 and a lighter Sprint 3 over the holidays.



proposed plan · kickoff Oct 5, 2026

MVP must-haves (cut line)

- [ ] Widget with auto-theme, Shopify + WordPress install
- [ ] RAG over help articles with sources
- [ ] Task engine with priority queue, step stack, undo stack, dead-letter queue
- [ ] Three actions: order status, return/exchange, address change
- [ ] Escalation queue + built-in agent inbox + Zendesk ticket fallback
- [ ] Admin dashboard: setup, knowledge, tasks, conversations, feedback inbox
- [ ] Thumbs, CSAT and agent handoff ratings
- [ ] Eval suite (300 cases) and red-team set in CI

Anything else waits for v1.1 unless it replaces a must-have.

Team

Role	People	Owns
Product manager (also business analyst)	1	PRD, backlog, beta customers, pricing
Tech lead / architect	1	Architecture, task engine design, code reviews
Frontend engineer	1	Widget, dashboard, agent inbox
Backend engineer	1	API, workers, integrations (Shopify, Zendesk)

Role	People	Owns
AI engineer	1	Prompts, RAG, planner, guardrails, evals
QA engineer (SDET)	1	Test plans, E2E, load tests, beta UAT
Designer + DevOps	0.5 + 0.5	Figma, design system / CI/CD, Terraform, monitoring

Sprint plan

Sprint	Dates	Goal	Main stories
0	Nov 9 to Nov 20	Foundation	Repo, CI/CD, Terraform dev env, skeleton widget + API, coding standards
1	Nov 23 to Dec 4	Answers questions	Knowledge ingestion, RAG with sources, widget chat + streaming, auto-theme v1
2	Dec 7 to Dec 18	Runs tasks	Priority queue, planner, execution + undo stacks, order-status action, eval set v1
3	Dec 21 to Jan 1	Hands off (reduced capacity)	Escalation queue, handoff packet, agent inbox, Zendesk fallback
4	Jan 4 to Jan 15	Ready to ship	Return/exchange + address actions, dashboard analytics, feedback inbox, load + security tests

Definition of done

- Merged via reviewed PR; CI green; coverage at least 80%
- Acceptance criteria verified by QA
- Eval scores equal or better for any AI change
- No new critical or high security findings
- Docs and runbook updated; demoed to the PM

10. Testing, CI/CD, deployment and operations

The quality system follows the HelpBot playbook (test pyramid, eval gates, canary releases) and adds two things the task engine needs: failure-injection tests for rollback and a widget compatibility matrix across real host sites.

Test plan

Test type	Tool	Example for this product	Runs
Unit	pytest, Vitest	Heap orders tasks by deadline; undo stack pops in reverse order	Every PR
Data-structure property tests	Hypothesis	Random step DAGs always execute in a valid topological order; no step runs twice	Every PR
Integration	pytest + Testcontainers	Worker pulls from Redis, writes task state to Postgres, survives restart	Every PR
Failure injection	Custom fault flags	Step 3 of an exchange fails: return label is cancelled, task escalated	Every PR touching the engine
AI evals	Promptfoo / Ragas + custom	300 questions and 60 multi-step tasks scored for correctness, groundedness, plan validity	PRs touching AI, nightly
Red team	Custom set	Prompt injection, other-customer data requests, abuse	Every AI PR + before release
Widget compatibility	Playwright on 20 real theme templates	Theme matches, no CSS bleed, no layout shift	Nightly
E2E	Playwright	Customer asks, sees progress card, gets handed to agent	After deploy to staging
Load	k6	1,000 concurrent chats; p95 under 3 s; queue lag under 1 s	Before release
Accessibility	axe + screen reader pass	Widget usable by keyboard and screen reader	Before release

Launch quality gate

- Correctness at least 90%, groundedness at least 98%, plan validity at least 95%, red-team refusal 100%
- Zero stuck tasks in a 24-hour soak test (every task reaches done, escalated or dead-letter)
- No open critical or high bugs; pen test and privacy review signed off
- Beta stores rate the product 4/5 or higher and 8 of 10 agree to convert to paid

CI/CD and environments

- GitHub Flow: short-lived `feature/SC-xx` branches, protected `main`, squash merges, conventional commits, release tags `vX.Y.Z`.
- Pipeline: lint, unit tests, SAST and secret scan, integration tests, quick eval (100 cases), build images, image scan, deploy to Dev, then Staging with full eval and E2E, then manual approval and canary to Production.
- Widget releases are versioned on the CDN; each tenant can be pinned to a version, and new versions roll out 5%, 25%, 100% by tenant.
- Environments: Local (docker compose), Dev, Staging (anonymized copies of beta-store data), Production.

Operations

Area	Metric	Alert when
Reliability	API error rate, worker crashes	Error rate over 1% for 5 min
Queues	Task queue depth and oldest-task age; escalation wait	Oldest task over 60 s; escalation wait over tenant SLA
Engine health	Rollback count, dead-letter count	Dead letters over 10 per hour
Speed	p95 latency, LLM latency	p95 over 3 s for 10 min
Cost	Tokens and spend per conversation, per tenant	Daily spend over 120% of budget
Quality	Thumbs-down rate, handoff rate, CSAT	Thumbs-down rate doubles week over week
Safety	Blocked injections, PII detections	Spike over baseline

On-call rotation with runbooks; blameless post-mortems; a kill switch per tenant turns the AI off so all chats go straight to humans.

11. Business model, go-to-market and KPIs

Flat monthly tiers by conversation volume, with unlimited agent seats and no per-resolution fee, undercut per-outcome competitors as soon as a store automates a few hundred chats a month. Prices below are proposals to test with beta stores.

Proposed pricing

Plan	Price / month	AI conversations	Includes
Free	$0	100	Widget, RAG answers, "Powered by" badge
Starter	$49	1,000	No badge, 1 action, agent inbox
Growth	$199	5,000	All actions, Zendesk handoff, analytics, feedback inbox
Scale	$599	20,000	Priority support, SSO, custom actions via webhook
Enterprise	Custom	Custom	SLA, data residency, dedicated model options

Overage: $15 per extra 1,000 conversations, with alerts at 80% and 100% and a hard cap the owner controls. The widget never goes dark: past the cap it falls back to human handoff.

Comparison point: a store with 2,000 AI resolutions a month pays about $1,980 in Fin outcome fees alone at $0.99 each, before seats (source). The same volume fits our $199 Growth plan.

Unit economics to prove in the PoC: LLM + infrastructure cost per conversation must stay under $0.03 so Growth keeps a gross margin above 70%. Levers: small model for routing, LRU answer cache, short prompts, larger model only for planning.

Go-to-market

1. **Design partners (now to launch):** recruit 10 beta stores from Shopify and WooCommerce communities; free for 3 months in exchange for weekly feedback calls.
2. **Marketplaces at launch:** Shopify App Store and WordPress plugin directory listings with a 10-minute setup video.

3. **Content:** honest pricing comparisons and a public "AI cost calculator" that shows per-resolution vs. flat pricing at the visitor's volume.
4. **Agencies:** revenue share for web agencies that install the widget on client stores.
5. **Product-led growth:** the free tier's badge links to a signup page; upgrade prompts appear when usage passes 80%.

KPIs

KPI	Definition	Target at 6 months post-launch
Automation rate	Chats resolved without a human, confirmed or no follow-up in 24 h	40%
Task success rate	Multi-step tasks completed without rollback or escalation	85%
CSAT	Average post-chat rating	4.3 / 5
Handoff wait	Median time from escalation to agent reply, business hours	Under 2 min
Wrong-answer rate	Thumbs-down tagged "wrong" per 100 AI answers	Under 2
Activation	Signups that install and get a first AI answer within 24 h	60%
Paying stores	Active paid tenants	50
Net revenue retention	Revenue kept and expanded from month-1 cohort	Over 100%
Cost per conversation	LLM + infra cost	Under $0.03

12. Risks, open questions and next steps

The biggest risks are AI taking a wrong action on a real order and LLM costs eating the flat-price margin; both are addressed in the PoC before the build is funded.

Risks

Risk	Impact	Mitigation
AI takes a wrong action (wrong order changed)	High	Ownership check before every action, customer confirmation for side effects, undo stack, action allow-list per tenant
LLM cost per conversation too high for flat pricing	High	Model routing, caching, per-tenant budgets; prove under $0.03 in PoC
Prompt injection via customer text or help articles	High	Input/output guardrails, tool calls validated against schema, red-team set in CI
Auto-theming produces unreadable colors	Medium	Contrast check with AA fallback; owner override
Widget breaks on unusual host sites	Medium	Shadow DOM, nightly compatibility matrix, per-tenant version pinning
Incumbents cut prices (e.g. Fin under Salesforce)	Medium	Compete on neutrality, task reliability and native look, not price alone
Customer distrust of AI support	Medium	AI disclosure, always-on human button, visible sources
Privacy and compliance (GDPR/CCPA)	Medium	PII masking, DPA with LLM provider, retention settings, data export/delete

Open questions

- [ ] Which LLM provider and models give the best cost-to-quality at our target of under $0.03 per conversation?
- [ ] Do beta stores accept flat tiers at $49 / $199 / $599, or do they want usage-only pricing?
- [ ] Is Shopify-first right, or should WooCommerce launch at the same time?
- [ ] Should the built-in agent inbox be a core product or only a fallback to existing helpdesks?
- [ ] How much of the task planner can be deterministic templates vs. LLM-generated plans for v1?

Next steps (first 2 weeks)

- [ ] Approve the discovery and PoC budget
- [ ] Interview 15 store owners and 10 support agents; collect 200 real anonymized chats

- [ ] Build a PoC: widget + RAG + one two-step task with rollback, measured on 50 real questions
- [ ] Measure cost per conversation and answer correctness in the PoC
- [ ] Sign 10 design-partner stores
- [ ] Go/No-Go review for the MVP build with the numbers above

Sources

- Grand View Research, Chatbot Market report (June 2026)
- MarketsandMarkets, AI for Customer Service Market
- Gartner, agentic AI to resolve 80% of common issues by 2029
- Gartner, half of companies that cut staff due to AI will rehire by 2027
- Gartner survey on customers preferring no AI in service (via BuildMVPFast)
- Macha, Intercom Fin pricing (checked Sep 24, 2026)
- Voiceflow, Zendesk AI agents pricing
- Chatarmin, Gorgias pricing 2026
- Chatarmin, Tidio pricing 2026
- Chatbase, product and pricing
- Reference design: "How a Big Company Builds an AI Project: HelpBot, End to End" (attached PDF)